Section 5.3 — Analysis

Adaptive Normalized Representation Learning for Generalizable Face Anti-Spoofing

Pattern Recognition — Final Slide

University of Science, VNU HCMC

Outline

5.3.1 Feature Map Analysis

- CNN feature map formulation
- High vs. low weight channels
- Adaptive feature combination

5.3.2 Balance Factor Analysis

- Layer-wise α behavior
- Sample-adaptive normalization
- Domain discrepancy effect

- Both analyses explain why AFNM generalizes better than fixed-normalization baselines.

Feature Map Analysis: CNN Feature Representation

Feature maps = intermediate activations capturing edges, textures, semantic structures.

A convolutional layer applies learnable filters to produce output feature maps:

$$F_k = \sigma(W_k * X + b_k)$$

W_k, b_k	weights and bias of the k-th filter
$*$	convolution operation
$\sigma(\cdot)$	non-linear activation
$X \in \mathbb{R}^{C \times H \times W}$	input feature tensor

In AFNM, each channel is normalized via a balance-weighted combination of BatchNorm and InstanceNorm:

$$\hat{x}_c = \underbrace{(1 - \alpha_c)}_{\text{IN weight}} \cdot \text{IN}(x_c) + \underbrace{\alpha_c}_{\text{BN weight}} \cdot \text{BN}(x_c)$$

The scalar α_c is learned per sample and channel.

Feature Map Analysis: High vs. Low Weight Channels

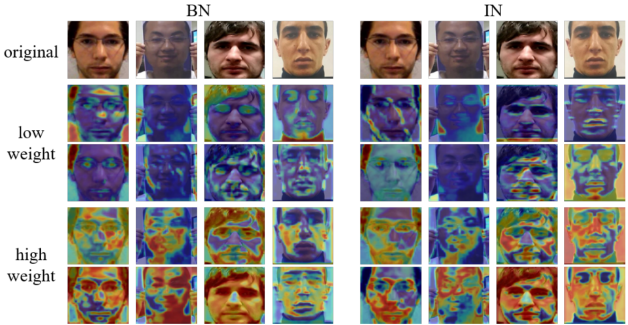


Figure: Feature maps of low/high-weight channels (IN and BN)

Key observations:

- High-weight channels (large α)
  - Focus on facial regions
  - Capture intrinsic spoofing cues
  - Transferable across domains
- Low-weight channels (small α)
  - Capture background / artifacts
  - Domain-specific noise

AFNM suppresses domain noise, preserves discriminative info $\Rightarrow$ better generalization.

Feature Map Analysis: Weighted Fusion

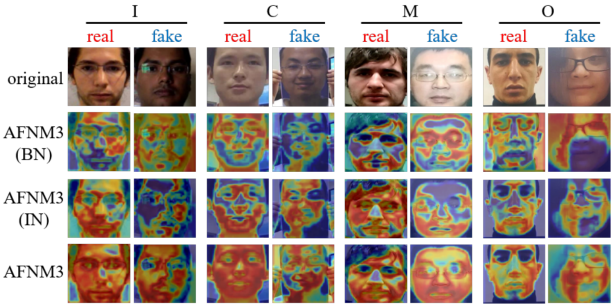


Figure: Weighted BN, IN channels and their fusion (AFNM3, task I&C&M → O)

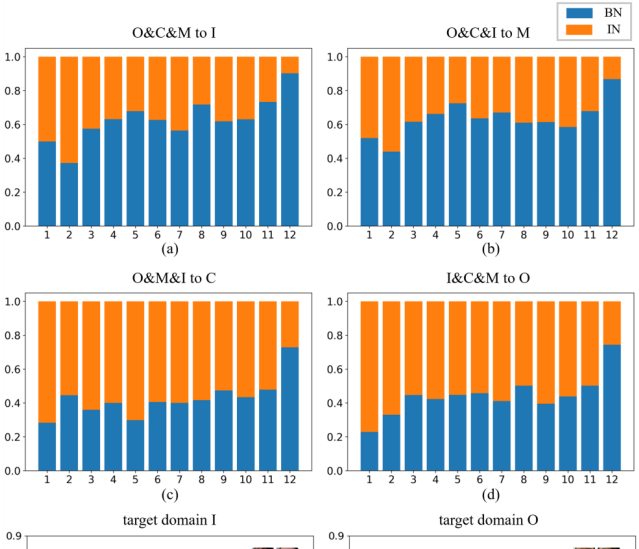
BN and IN channels are combined per channel with learned weights.

Why fusion works

- BN captures class-discriminative features
- IN removes domain-specific variations
- Adaptive weighting selects the right balance per sample

The fused representation is less sensitive to domain shift while retaining spoofing-sensitive patterns.

Balance Factor α: Layer-wise Behavior



α controls BN vs. IN contribution:

Shallow layers (L_1, L_2)

$\alpha \approx 0$ small

IN dominates $\Rightarrow$ removes domain-specific variations

Deep layers (L_3, L_4)

$\alpha \to 1$ large

BN dominates $\Rightarrow$ extracts high-level discriminative features

Task with larger domain discrepancy $\Rightarrow$ more IN reliance (α lower overall).

Balance Factor α: Sample Adaptation and Feature Geometry

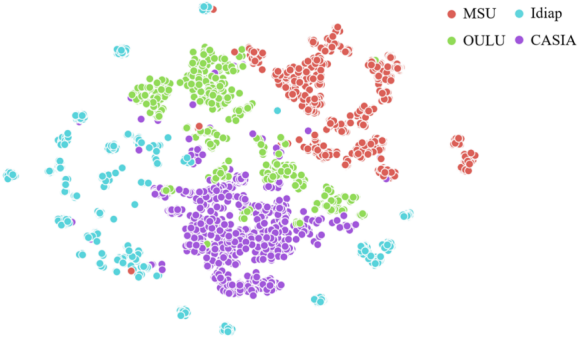


Figure: t-SNE visualization of features from ResNet (ImageNet pretrained)

Sample-adaptive normalization:

- α varies significantly across samples
- Model adjusts strategy per input
- Handles diverse attack types and domain shifts

t-SNE insight:

- Real vs. fake samples are more separable with AFNM
- Feature clusters are more compact across domains

Confirms adaptive normalization improves cross-domain discriminability.

Summary: Key Takeaways from Section 5.3

1. **Feature maps** from CNNs encode hierarchical patterns; AFNM adaptively weights channels to suppress domain-specific noise while preserving spoofing cues.
2. **High-weight channels** (BN+IN) focus on facial regions; **low-weight channels** capture background / artifacts — AFNM suppresses the latter.
3. **Balance factor** α is layer-dependent: shallow $\Rightarrow$ IN (remove domain variation), deep $\Rightarrow$ BN (extract discriminative features).
4. α is sample-adaptive — the model tailors normalization per input, which is crucial for handling diverse attack types and domain shifts.
5. t-SNE confirms AFNM features are more separable and compact across domains than fixed-normalization baselines.

$$\boxed{\text{Adaptive normalization} \Rightarrow \text{robust domain generalization for face anti-spoofing.}}$$